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1

Integers



In this chapter you will learn about...

- ☐ Integers
- ☐ Addition and subtraction of integers and their properties
- ☐ Multiplication of integers and its properties
- ☐ Division of integers and its properties

INTEGERS

In whole numbers $\{0, 1, 2, 3, 4, 5, 6, 7, 8, \dots\}$, we can subtract a smaller number from a larger number but it is not possible to subtract a larger number from a smaller number.

$$7 - 3 = 4; \quad 8 - 5 = 3; \quad 5 - 2 = 3;$$

$$3 - 7 = ?; \quad 5 - 8 = ?; \quad 2 - 5 = ?$$

So, a new type of numbers were introduced denoted by $-1, -2, -3, -4, -5, -6, \dots$ such that $1 + (-1) = 0, 2 + (-2) = 0, 3 + (-3) = 0$ and so on

Thus, we get numbers as

$$\dots, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, \dots$$

These numbers are called **Integers**.

The numbers $1, 2, 3, 4, \dots$ are called **positive integers**.

The numbers $-1, -2, -3, -4, \dots$ are called **negative integers**.

0 is simply an integer, neither positive nor negative.

The integers can be represented on the number line as given below :



The arrows on both sides of the number line indicate that the number line can be extended infinitely in both directions.

Look at the number line carefully and observe :

(a) A number on the number line is greater than every number on its left.

Examples : (i) $3 > 2$ as 3 is to the right of 2.

(ii) $0 > -1$ as 0 is to the right of -1.

(iii) $-2 > -3$ as -2 is to the right of -3.

(b) If 'a' and 'b' are two integers and $a > b$, then $-a < -b$.

Examples : (i) $3 > 1 \Rightarrow -3 < -1$

(ii) $7 > 5 \Rightarrow -7 < -5$

- 7 is additive inverse of 7 or 7 is additive inverse of - 7.

Example : $7 + (-7) = 0 = -7 + 7$

- a is called the additive inverse of a.

$$a + (-a) = 0 = -a + a$$

5. **Additive Inverse :** If a is any integer, then there exists an integer - a such that

Hence, 0 is called the additive identity.

$$(ii) (-7) + 0 = 0 + (-7) = -7$$

Examples : (i) $3 + 0 = 0 + 3 = 3$;

4. **Additive Identity :** If a is any integer, then, $a + 0 = 0 + a = a$.

∴ In addition of integers, order does not matter.

Hence, $(-6) + 7 + (-9) = (-6) + (7 + (-9))$

and $(-6) + (7 + (-9)) = -6 + (-2) = -(6 + 2) = -8$

Example : $(-6) + 7 + (-9) = 1 + (-9) = -8$

$$(a + b) + c = a + (b + c)$$

3. **Associative Property :** If a, b and c are integers, then

∴ The sum of the integers remains the same in whichever order we add the integers.

Hence, $(-9) + (-5) = (-5) + (-9)$

$$(ii) (-9) + (-5) = -(9 + 5) = -14 \text{ and } (-5) + (-9) = -(5 + 9) = -14$$

Hence, $6 + (-11) = (-11) + 6$

Examples : (i) $6 + (-11) = -5$ and $(-11) + 6 = -5$

$$a + b = b + a$$

2. **Commutative Property :** If a and b are integers, then

∴ The integers are closed under addition.

$$(iii) -9 + 13 = (13 - 9) = 4, \text{ which is an integer.}$$

$$(ii) -4 + (-3) = -(4 + 3) = -7, \text{ which is an integer.}$$

Examples : (i) $-8 + 5 = -(8 - 5) = -3$, which is an integer.

1. **Closure Property :** If a and b are integers, then a + b is also an integer.

Since, all whole numbers are integers, therefore, all properties of operations on whole numbers remain useful. We sum up the properties as follows :

PROPERTIES OF ADDITION OF INTEGERS

$$= -(9 + 13) = -22$$

$$(ii) (-9) - (13) = -9 + (-13) \text{ (negative of 13 is -13)}$$

$$= -(15 - 8) = -7$$

Solution : (i) $(-15) - (-8) = -15 + 8$ (negative of -8 is 8)

Example 3 : Subtract : (i) (-8) from (-15). (ii) 13 from (-9).

and then add them.

Rule : To subtract one integer from the other integer, convert the integer to be subtracted to its negative

SUBTRACTION OF INTEGERS

$$(ii) \quad (-13) + (+8) = -(13-8) = -5$$

Hence, $(+15) + (-7) = +8$

Example 2 : Add : (i) 15 and (-7) .
(ii) (-13) and 8.

Rule 2 : To add a positive and a negative integer, we find the difference between their numerical values regardless of their signs and put the sign of the integer with the greater value to it.

$$(ii) \quad (-8) + (-5) = -(8+5) = -13$$
$$(+15) + (+8) = (+15 + 8) = +23$$

Example 1 : Add : (i) $(+15)$ and $(+8)$.
(ii) (-8) and (-5) .

common sign with the sun.

Rule 1 : To add two integers of like signs, we add their values regardless of their signs and put the

ADDITION OF INTEGERS

Hence, $-|-7| - |-5| = -12$

$$= -7 - 5 = -12$$
$$|(-5) - (-7)| - |(-7) - (-5)| = |-5| - |-7| = \quad (ii)$$

Hence, $|-9| - |3| = 6$

$$9 = 3 - 6 =$$
$$3 - (6 -) = |3| - |6 -| \quad (iii)$$

Hence, $|-3| + |-2| = 5$

$$5 = 3 + 2 = 5$$
$$|-3| + |-2| = -(-3) + -(-2) \quad (ii)$$

Hence, $|7| + |-7| = 14$

$$= 7 + 7 = 14$$

Solution : (i) $|7| + |-7| = |-7| + |7|$

$$(i) \quad |7| + |-7| \quad (ii) \quad |-3| + |-2| \quad (iii) \quad |-9| - |3| \quad (iv) \quad -|-7| - |-5|$$

Example 1 : Simplify :

$$\left. \begin{array}{l} \text{if } x \text{ is positive or zero} \\ \text{if } x \text{ is negative} \end{array} \right\} \begin{array}{l} x \\ -x \end{array} = |x|$$

Thus,

$$|-5| = 5, \quad |5| = 5$$

For example :

The numerical value of an integer regardless of its sign is called its absolute value. The absolute value of an integer is always positive or zero. We denote it by $|x|$.

ABSOLUTE VALUE

EXERCISE 1.1



1. Find the sum of :
 - (i) 256 and - 312
 - (ii) - 319 and - 136
 - (iii) - 127 and 300
 - (iv) - 197 and 214
 - (v) - 84 and - 112
 - (vi) 215 and - 42
2. Subtract :
 - (i) 32 from - 56
 - (ii) - 73 from - 80
 - (iii) - 63 from 72
 - (iv) - 32 from 0
 - (v) - 92 from - 23
 - (vi) - 26 from 59
3. Subtract the sum of - 524 and 678 from - 92.
4. Subtract - 341 from the sum of 83 and - 56.
5. The sum of two integers is - 278. If one of the integers is - 156, find the other.
6. Subtract the sum of - 42 and - 147 from the difference of - 283 and - 415.
7. Simplify : $14 - (-18) + [-32 - (-15)]$
8. If $a = -10$, $b = -4$, show that $a - b \neq b - a$.
9. Find the value of :
 - (i) $3 + (-4) + (-5) - (-4)$
 - (ii) $(-6) + (-9) + (-41)$
10. Simplify :
 - (i) $[23 - (-9)] + [12 - (-6)]$
 - (ii) $-15 + (-63) - (-45) + (-16)$

MULTIPLICATION OF INTEGERS

In order to multiply integers, we follow the following rules :

- Rule 1 :** The product of two integers with like signs is equal to the product of their absolute values with positive sign.
- Examples :**
- (i) $4 \times 6 = 24$, $(+ve) \times (+ve) = (+ve)$
 - (ii) $(-5) \times (-4) = 20$, $(-ve) \times (-ve) = (+ve)$
 - (iii) $(-8) \times (-9) = 72$, $(-ve) \times (-ve) = (+ve)$
- Rule 2 :** The product of two integers of opposite signs is equal to the product of their absolute values with negative sign.
- Examples :**
- (i) $4 \times (-9) = -36$, $(+ve) \times (-ve) = (-ve)$
 - (ii) $(-8) \times 7 = -56$, $(-ve) \times (+ve) = (-ve)$

Example 1 : Evaluate :

- (i) $(-15) \times 8$
- (ii) $9 \times (-12)$
- (iii) $(-7) \times (-9)$
- (iv) 21×6
- (v) $(-13) \times (-5)$
- (vi) $11 \times (-5)$
- (i) $(-15) \times 8 = -120$
- (ii) $9 \times (-12) = -108$
- (iii) $(-7) \times (-9) = 63$
- (iv) $21 \times 6 = 126$
- (v) $(-13) \times (-5) = 65$
- (vi) $11 \times (-5) = -55$

Solution :

PROPERTIES OF SUBTRACTION OF INTEGERS

1. Closure Property : If a and b are two integers, then $a - b$ is also an integer.

Examples : (i) $11 - 5 = 6$
 (ii) $(-13) - 2 = -(13 + 2) = -15$
 (iii) $(-15) - (-8) = -15 + 8 = -7$
 ∴ The integers are closed under subtraction.
2. Commutative Property : If a and b are integers, then

Examples : (i) $3 - 5 = -2$, $5 - 3 = 2$
 Hence, $3 - 5 \neq 5 - 3$
 (ii) $(-4) - (-7) = -4 + 7 = 3$,
 $(-7) - (-4) = -7 + 4 = -3$
 Hence, $(-4) - (-7) \neq (-7) - (-4)$
 ∴ Commutative property of subtraction does not hold true in integers.
3. Associative Property : If a , b and c are integers, then

Example :
 $(a - b) - c \neq a - (b - c)$
 $(9 - 3) - 4 = 6 - 4 = 2$ and
 $9 - (3 - 4) = 9 - (-1) = 9 + 1 = 10$
 ∴ $(9 - 3) - 4 \neq 9 - (3 - 4)$
 ∴ Associative property of subtraction does not hold true in integers.

Example 4 : Evaluate the following :

- (i) $13 + (-7)$
- (ii) $(-17) + 6$
- (iii) $(-7) + (-13)$

Solution :

$$(i) \quad 13 + (-7) = 13 - 7 = 6$$

$$(ii) \quad (-17) + 6 = -(17 - 6) = -11$$

$$(iii) \quad (-7) + (-13) = -(7 + 13) = -20.$$

Example 5 : Find the additive inverse of

- (i) -72
- (ii) 325
- (iii) 1
- (iv) -3456

Solution :

∴ The additive inverse of -72 is 72 .

(ii) The opposite of integer 325 is -325 , such that $325 + (-325) = 0$.

∴ The additive inverse of 325 is -325 .

(iii) The opposite of integer 1 is -1 , such that $1 + (-1) = 0$.

∴ The additive inverse of 1 is -1 .

(iv) The opposite of integer -3456 is $+3456$, such that $-3456 + 3456 = 0$.

∴ The additive inverse of -3456 is 3456 .

(ii) Multiplicative inverse of (-8) is $-\frac{1}{8}$.

$$\therefore (-8) \times \left(-\frac{1}{8}\right) = 1$$

Example 2 : Verify : $17 \times [7 + (-3)] = (17 \times 7) + [17 \times (-3)]$

Solution :

$$\text{L.H.S.} = 17 \times [7 + (-3)]$$

$$= 17 \times 4 = 68$$

$$\text{R.H.S.} = (17 \times 7) + [17 \times (-3)]$$

$$= 119 + (-51) = 68$$

$$\text{Hence, } 17 \times [7 + (-3)] = (17 \times 7) + [17 \times (-3)]$$

Example 3 : Find the product, using suitable properties.

$$(i) \quad 15 \times (-25) \times (-4) \times (-10) \quad (ii) \quad 625 \times (-35) + (-625) \times 65$$

Solution :

$$(i) \quad 15 \times (-25) \times (-4) \times (-10)$$

$$= [15 \times (-10)] \times [(-25) \times (-4)]$$

$$= (-150) \times 100 = -15000.$$

$$(ii) \quad 625 \times (-35) + (-625) \times 65$$

$$= 625 [(-35) + (-65)]$$

$$= 625 \times (-100) = -62500$$

Example 4 : Using distributive property, find

$$(i) \quad 49 \times 17 \quad (ii) \quad (-26) \times (-41) \quad (iii) \quad (-39) \times (-15) + (-1) \times 39$$

Solution :

$$(i) \quad 49 \times 17 = (50 - 1) \times 17$$

$$= 50 \times 17 - 1 \times 17$$

$$= 850 - 17 = 833$$

$$\text{Hence, } 49 \times 17 = 833$$

$$(ii) \quad (-26) \times (-41) = 26 \times 41$$

$$= 26 \times (40 + 1)$$

$$= 26 \times 40 + 26 \times 1$$

$$= 1040 + 26 = 1066$$

$$\text{Hence, } (-26) \times (-41) = 1066$$

$$(iii) \quad (-39) \times (-15) + (-1) \times 39$$

$$= (-39) \times [(-15) + 1]$$

$$= (-39) \times (-14)$$

$$= 39 \times 14$$

$$= (40 - 1) \times 14$$

$$= 40 \times 14 - 1 \times 14$$

$$= 560 - 14 = 546$$

Example 5 : Find each of the following products :

$$(i) \quad 9 \times (-2) \times (-5)$$

$$(iii) \quad (-11) \times (-12) \times 10$$

$$(iv) \quad (-3) \times (-7) \times (-6)$$

$$(ii) \quad (-4) \times (-6) + (-3) \times (-1)$$

$$[\therefore (-ve) \times (-ve) = (+ve)]$$

$$[\therefore (-ve) \times (-ve) = (+ve)]$$

Properties of Multiplication of Integers

1. **Closure Property :** If a and b are integers, then $a \times b$ is an integer.

Examples : (i) $6 \times 5 = 30$, which is an integer.

(ii) $8 \times (-6) = -48$, which is an integer.

(iii) $(-4) \times 9 = -36$, which is an integer.

(iv) $0 \times (-7) = -(0 \times 7) = 0$, which is an integer.

∴ Integers are closed under multiplication.

2. **Commutative Property :** If a and b are two integers, then

$$a \times b = b \times a$$

Example :

$$(-6) \times 7 = -(6 \times 7) = -42 \text{ and } 7 \times (-6) = -(7 \times 6) = -42$$

Hence,

$$(-6) \times 7 = 7 \times (-6)$$

∴ The product of two integers remains the same whatever be the order of multiplication of integers.

3. **Associative Property :** If a , b and c are three integers, then

$$a \times (b \times c) = (a \times b) \times c$$

Example :

$$7 \times [(-4) \times (-2)] = 7 \times 8 = 56$$

and

$$[7 \times (-4)] \times (-2) = (-28) \times (-2) = 56$$

Hence,

$$7 \times [(-4) \times (-2)] = [7 \times (-4)] \times (-2)$$

The product of three integers remains the same whatever be the order of multiplication. ∴ The multiplication is associative for integers.

4. **Multiplicative Identity :** If a is any integer, then

$$a \times 1 = 1 \times a = a$$

Examples :

$$(i) \quad 12 \times 1 = 1 \times 12 = 12$$

$$(ii) \quad (-5) \times 1 = 1 \times (-5) = -5$$

1 is called the multiplicative identity.

5. **Property of Zero :** If a is any integer, then $a \times 0 = 0 \times a = 0$.

Examples :

$$(i) \quad 0 \times 6 = 6 \times 0 = 0$$

$$(ii) \quad (-5) \times 0 = 0 \times (-5) = 0$$

6. **Distributive Property of Multiplication Over Addition :** If a , b and c are three integers, then

$$a \times (b + c) = (a \times b) + (a \times c)$$

Example :

$$(-4) \times [(-5) + 3] = (-4) \times (-2) = 8$$

and

$$[(-4) \times (-5)] + [(-4) \times 3] = 20 + (-12) = 8$$

Hence,

$$(-4) \times [(-5) + 3] = [(-4) \times (-5)] + [(-4) \times 3]$$

7. **Multiplicative Inverse :** Multiplicative inverse of non-zero integer a is $\frac{1}{a}$.

$$a \times \frac{1}{a} = \frac{1}{a} \times a = 1$$

Examples : (i) Multiplicative inverse of 7 is $\frac{1}{7}$.

$$7 \times \frac{1}{7} = 1$$

DIVISION OF INTEGERS

Division is inverse of multiplication.

- (i) Dividing 18 by 3 implies that we have to find an integer which when multiplied by 3 gives 18.

Obviously, such an integer is 6.

$$18 \div 3 = \frac{18}{3} = 6, \text{ as } 3 \times 6 = 18$$

(ii) Similarly, $42 \div (-7) = \frac{42}{-7} = -6$, as $-7 \times (-6) = 42$

Let us recall few definitions relating to division :

Dividend : The number to be divided is called the

dividend.

Divisor : The number which divides is called the divisor.

Quotient : The result of division is called the quotient.

Remainder : The number that is left over in division is

called the remainder.

To find the quotient of two integers we have the following rules:

Rule 1 : If two integers are both positive or both negative, then absolute value of two integers are divided and positive sign is assigned to the quotient.

Examples : (i) $12 \div 4 = 3$ (ii) $(-42) \div (-7) = 42 \div 7 = 6$

Rule 2 : If two integers are of different signs, then absolute value of two integers are divided and negative sign is assigned to the quotient.

Examples : (i) $(-26) \div 2 = -13$, (ii) $24 \div (-6) = -4$

Properties of Division of Integers

1. **Closure property :** If a and b are integers, then $a \div b$ is not always an integer.

Examples : (i) $(-14) \div 2 = -7$, which is an integer

(ii) $(-15) \div (-3) = 5$, which is an integer

(iii) $(-7) \div 2$ is not an integer

(iv) $(-13) \div 3$ is not an integer

2. If a is an integer such that $a \neq 0$, then $a \div a = 1$ and $a \div 1 = a$.

Examples : $7 \div 7 = 1$; $(-8) \div (-8) = 1$, $(-9) \div 1 = (-9)$

3. For any non-zero integer a , $0 \div a = 0$.

Examples : $0 \div 6 = 0$, $0 \div (-6) = 0$

4. If a is an integer, then $a \div 0 = \frac{a}{0}$ is not defined.

Example 1 : Find the quotient in each of the following :

(i) $(-63) \div (-9)$ (ii) $1078 \div (-11)$ (iii) $(-12096) \div 576$

Solution : (i) $\frac{(-63)}{(-9)} = \frac{63}{9} = 7$

$ \begin{array}{r} 20 \div 3 = \frac{20}{3} \\ 6 \leftarrow \text{Quotient} \\ \underline{18} \\ 20 - 18 \\ \underline{2} \\ 20 - 18 \\ \underline{2} \end{array} $	Divisor $\rightarrow$ 3 Dividend $\rightarrow$ 20 Remainder $\rightarrow$ 2 Quotient $\leftarrow$ 6 Dividend $\leftarrow$ 20 Remainder $\leftarrow$ 2
---	--

8. If '*' operation be defined as $a * b = a \times b - (a + b)$, evaluate :
- (i) $2 * 5$ (ii) $(-4) * (-6)$ (iii) $(-13) * (-6)$
- (iii) $b \times c = c \times b$
- (i) $b \times (a + c) = b \times a + b \times c$ (ii) $b \times c \times a = a \times b \times c = c \times a \times b$
7. If $a = 3$, $b = -2$, $c = -1$, verify that :
- (i) $(-5) \times 19 \times (-60)$ (ii) $25 \times 31 \times (-4)$
- (iii) $97 \times 25 \times (-2) \times (-4)$ (iv) $(-2) \times 4896 \times (-5)$
6. Using convenient grouping, evaluate :
- (i) 23 (ii) -47 (iii) 0
5. Determine the integer whose product with '-1' is
- (i) 6 (ii) $\frac{3}{-1}$ (iii) -4 (iv) $-\frac{5}{2}$
4. Find the multiplicative inverse of the following :
- (i) $23 \times 17 + (-2) = 23 \times 7 + 23 \times (-2)$
- (ii) $(-18) \times [(-3) + (-5)] = [(-18) \times (-3)] + [(-18) \times (-5)]$
3. Verify the following :
- (i) $3276 \times 3 - 3276 \times 2$ (ii) $8964 \times 1 + 99 \times 8964$
- (iii) $1695 \times 678 - 695 \times 678$ (iv) $2314 \times 9 + 2314$
2. Find the value of
- (i) $6 \times (-12)$ (ii) $10 \times (-6) \times (-1)$
- (iii) $(-1) \times (-5) \times (-7) \times (-2)$ (iv) $6 \times (-5) \times (-5) \times (-2)$
- (v) $20 \times (-123) \times (-5)$ (vi) $(-12) \times (-5) \times 12$
- (x) $(-6) \times (-3) \times (-1) \times (-2)$ (ix) $3 \times (-8) \times 5$
1. Find each of the following products :

EXERCISE 1.2



- Solution :
- (i) $9 \times (-2) \times (-5) = 9 \times [(-2) \times (-5)]$
 $= 9 \times 10 = 90$
- (ii) $(-4) \times (-6) + (-3) \times (-1) = [(-4) \times (-6)] + [(-3) \times (-1)]$
 $= 24 + 3 = 27$
- (iii) $(-11) \times (-12) \times 10 = [(-11) \times (-12)] \times 10$
 $= (11 \times 12) \times 10$
 $= 132 \times 10 = 1320$
- (iv) $(-3) \times (-7) \times (-6) = [(-3) \times (-7)] \times (-6)$
 $= (3 \times 7) \times (-6)$
 $= 21 \times (-6) = -126$
- $[\therefore (-ve) \times (-ve) = (+ve)]$
- $[\therefore (-ve) \times (-ve) = (+ve)]$

Word Problems

Example 1 : In a class test containing 10 questions, 4 marks are given for every correct answer and (-2) marks are given for every incorrect answer.

- (i) Naresh attempts all questions but only 8 of his answers are correct. What is his total score?
 (ii) Umesh also attempts all questions but gets only 6 answers correct. What will be his score?

Solution : (i) Correct answers = 8

Marks given for 1 correct answer = 4

Marks given for 8 correct answers = $8 \times 4 = 32$

Marks given for 1 incorrect answer = -2

Marks given for 2 incorrect answers = $2 \times (-2) = -4$

Total score of Naresh = $32 + (-4) = 28$

$\therefore$ Total score of Naresh is 28.

(ii) Marks given for 1 correct answer = 4

Marks given for 6 correct answers = $6 \times 4 = 24$

Marks given for 1 incorrect answer = -2

Marks given for 4 incorrect answers = $4 \times (-2) = -8$

Total score of Umesh = $24 + (-8) = 16$

$\therefore$ Total score of Umesh is 16.

Example 2 : The product of two integers is -182. If one of the integers is 13, find the other integer.

Solution : Product of two integers = -182

One integer = 13

$\therefore$ Other integer = $(-182) \div 13$

$$= \frac{-182}{13}$$

$$= -14$$

The other integer is -14.

Example 3 : Find an integer which when multiplied by 3 and then divided by 2 becomes (-18).

Solution : Integer multiplied by 3 and then divided by 2 = -18

$$\Rightarrow \text{Integer multiplied by } \frac{3}{2} = -18$$

$$\Rightarrow \text{Integer} \times \frac{3}{2} = -18$$



A man wanted to get into his work building. But he had forgotten his codes. However he did remember five clues. The fifth digit plus the third digit equals 14. The fourth digit is one more than the second digit. The first digit is one less than twice the second digit. The second digit plus the third digit equals ten. The sum of all five digits is 30. What are the five digits and in what order?

- an integer
6. Give three examples to prove that, if p and q are two integers, then $p + q$ is not necessarily an integer
5. Write four pairs of integers (a, b) such that $a + b = -4$.
4. If $a = 24, b = 8, c = 4$, verify that

3. Simplify :

(i) $(-441) - 21$

(ii) $(-49) + 7 + (-7)$

(iii) $(45 + 9) - [(-36) + 9]$

(iv) $[(-56) + (-8)] + [(-7) + 7]$

(v) $[(-11) + (-5)] + [3 + (-1)]$

2. Write each of the following in simplest form :

(i) $45 \div (-9)$

(ii) $(-243) \div 9$

(iii) $(-13) - 195$

(iv) $(-5525) + (-221)$

(v) $1728 \div (-12)$

(vi) $729 \div 9$

(vii) $(-133) + (-19)$

(viii) $(-90) + (-15)$

(ix) $105 + (-21)$

(x) $(-810) + 27$

1. Find the quotient in each of the following :
- (i) $(-36) \div (-4)$
- (ii) $(-36) \div (-4)$
- (iii) $(-36) \div (-4)$
- (iv) $(-36) \div (-4)$
- (v) $(-36) \div (-4)$
- (vi) $(-36) \div (-4)$
- (vii) $(-36) \div (-4)$
- (viii) $(-36) \div (-4)$
- (ix) $(-36) \div (-4)$
- (x) $(-36) \div (-4)$

EXERCISE 1.3

Hence, (-288) should be divided by (-8) to get 36.

$\Rightarrow$

Integer = $36 \times (-8)$

$= -288$

Solution : Here, quotient = 36, divisor = -8 to get 367

Now, Integer + $(-8) = 36$

Example 3 : What should be divided by -8 to get 36?

(i) $(-42) + (-6) = -48$

(ii) $(-42) + (-6) = -48$

(iii) $(-42) + (-6) = -48$

(iv) $(-42) + (-6) = -48$

(v) $(-42) + (-6) = -48$

(vi) $(-42) + (-6) = -48$

(vii) $(-42) + (-6) = -48$

(viii) $(-42) + (-6) = -48$

(ix) $(-42) + (-6) = -48$

(x) $(-42) + (-6) = -48$

Solution :

(i) $(-24) + (-6) = -30$

(ii) $(-42) + (-6) = -48$

(iii) $(-32) + (-6) = -38$

(iv) $(-32) + (-6) = -38$

(v) $(-32) + (-6) = -38$

(vi) $(-32) + (-6) = -38$

(vii) $(-32) + (-6) = -38$

(viii) $(-32) + (-6) = -38$

(ix) $(-32) + (-6) = -38$

(x) $(-32) + (-6) = -38$

Example 2 : Simplify :

(i) $\frac{1078}{(-11)} = -\left(\frac{1078}{11}\right) = -98$

(ii) $\frac{-12096}{576} = -\left(\frac{12096}{576}\right) = -21$

Laughter Time



Teacher : What is $3 + 37$

Student : Six

Teacher : That's good.

Student : Good! That's perfect.

1. On subtracting -15 from -9 , we get
 - (a) -6
 - (b) -24
 - (c) 6
 - (d) 24
2. $3495 + 3495 \times 9 =$
 - (a) 3495
 - (b) 34950
 - (c) 6990
 - (d) none of these
3. $(-12) \times 6 - (-12) \times 4 =$
 - (a) 24
 - (b) 60
 - (c) -60
 - (d) -24
4. On subtracting (-8) from 6 , we get
 - (a) 2
 - (b) 14
 - (c) -14
 - (d) -2
5. The sum of two integers is -14 . If one of them is 20 , then the other is
 - (a) -34
 - (b) 6
 - (c) -6
 - (d) 34

Choose the correct alternatives in each of the following :

MULTIPLE CHOICE QUESTIONS



- The numbers $\dots, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, \dots$ are integers.
- 0 is an integer which is neither positive nor negative.
- ' 0 ' is less than every positive integer and greater than every negative integer.
- The absolute value of an integer is its numerical value regardless of its sign.
- To add two integers with like signs, we add their numerical values and put the common sign with the sum.
- To add two integers with unlike signs, we take the difference of their numerical values and put the sign of the integer having the greater absolute value.
- To subtract an integer b from an integer a , we change the sign of b and add it to a .
- $-a$ and a are additive inverse or negatives of each other.
- To find the product of two integers with like signs, we multiply their numerical values and give a plus sign to the product.
- To find the product of two integers with unlike signs, we multiply their numerical values and give a minus sign to the product.
- The quotient of two negative or two positive integers is always positive.
- The quotient of one positive and one negative integer is always negative.

FACTS TO REMEMBER



- (i) sum is - 9
(ii) difference is - 7
(iii) sum is 0

8. Write down a pair of integers whose

7. By what number should - 324 be divided to obtain - 18?

amount in the bank after withdrawal.

₹ 1500 and next day, he had to withdraw $\frac{1}{3}$ rd of his total balance. Find his balance

6. Rajan had ₹ 6000 in his bank account on a particular day. A week later, he deposited

starts from 20 m above the ground level, how long will it take to reach - 340 m?

5. An elevator descends into a mine shaft at the rate of 6 metres per minute. If the descent

40 pens. How many pencils did he sell in that period?

4. A shopkeeper earns a profit of ₹ 1 by selling one pen and incurs a loss of 30 paise on sale of one pencil. In a particular month, he incurs a loss of ₹ 5. In that month, he sold

3. If the quotient obtained on dividing an integer by - 9 is - 8, find the integer.

2. Find an integer which when multiplied by 4 and then divided by 9 becomes (- 28).

1. The product of two integers is 270. One of the integers is (- 18). Find the other integer.

EXERCISE 1.4



Each of her friends gets 4 apples.

Each will get = $12 \div 3 = 4$

3 friends get = 12 apples

Number of apples she distributed = $25 - 13 = 12$

Solution : Total apples Priyanka bought = 25

of her friends get?

Example 4 : Priyanka bought 25 apples. She kept 13 apples with her and distributed the remaining apples equally among her three friends. How many apples did each

The required integer is (- 12).

$$= \frac{-36}{3} = -12$$

$$\Rightarrow \text{Integer} = \frac{(-18) \times 2}{3}$$

VALUE CORNER

On her birthday, Nikita had 60 balloons. She kept 12 balloons and distributed the remaining equally among her 8 friends.

(a) How many balloons did each of her friends get?

(b) What value does Nikita show by this gesture?

$$(ii) (-5) \times (-13) \times (-1) \times 0 \times (-6)$$

1. Simplify : $|42 + (-7)| + |(-48) + 16|$
2. State, which is greater?
3. Find the value of :
 - (i) $(6 - 8) \times 10$ and $6 - (8 \times 10)$
 - (ii) $(12 + 6) \times 10$ and $12 + (6 \times 10)$
 - (iii) $(-3) - 7$ and $(-3) - (7 \times (-2))$
4. Write down a pair of integers whose
 - (i) sum is -5
 - (ii) difference is -9
 - (iii) product is -15
 - (iv) quotient is -7
5. A plane is flying at a height of 4500 m above sea level. At a particular point, it is exactly above a submarine floating 1100 m below sea level. What is the vertical distance between them?
6. Subtract -8 from 15 and subtract 15 from -8 . Are the two results same?
7. Evaluate :
 - (i) $-15 + (-23) - (-24) + (-17)$
 - (ii) 19 negative integers and 19 positive integers?
 - (iii) 12 negative integers and 9 negative integers?
 - (iv) 17 positive integers and 6 negative integers?
8. What will be the sign of the product, if we multiply together
 - (i) 5 negative integers and 3 positive integers?
 - (ii) 19 negative integers and 19 positive integers?
 - (iii) 12 negative integers and 9 negative integers?
 - (iv) 17 positive integers and 6 negative integers?

REVIEW EXERCISE



8. The negative of a negative integer is always a integer.
9. should be divided by -6 to get 25.
10. is the additive identity whereas is the multiplicative identity for integers.
11. For any integer a , $a + 1 =$ whereas $a + 0$ is
12. -12 is than -10 .

- Fill in the blanks :
- When two integers of like signs are multiplied, the product is a
 - The product of 7 negative integers is
 - The quotient of two integers with like signs is
 - The successor of -175 is
 - The additive inverse of 7 is
 - If a is any integer, then its predecessor is
 - The sum of two integers is 47 and one integer is -13 , the other integer is

MENTAL MATHS CORNER



- On subtracting 7 from -6 , we get
(a) 1 (b) -13 (c) 13 (d) -1
- If a, b, c are integers, then $(a + b) + c \neq a + (b + c)$ unless $c =$
(a) 1 (b) -1 (c) 0 (d) a
- 2 exceeds -3 by
(a) 1 (b) -1 (c) 5 (d) -6
- On subtracting 6 from -6 , we get
(a) -12 (b) 0 (c) 12 (d) none of these
- The smallest integer is
(a) 0 (b) 1 (c) -1 (d) not defined
- $0 + (-3)$ is equal to
(a) -3 (b) 0 (c) 3 (d) not defined
- $(-7) + 0$ is
(a) -7 (b) 7 (c) 0 (d) not defined
- On subtracting -5 from -7 , we get
(a) -12 (b) -2 (c) 2 (d) 12
- On multiplying largest three digit integer with the smallest two digit positive integer, we get
(a) 10001 (b) 9990 (c) 9900 (d) 9991
- What should be multiplied by -23 to get 575?
(a) 15 (b) 25 (c) -25 (d) 35

MATHS LAB ACTIVITY

Objective : To prove multiplication of integers is commutative.
Activity : Complete the following table :

S. No.	Statement-1	Statement-2	Inference
1.	$2 \times (-4) = -8$	$(-4) \times 2 = -8$	$2 \times (-4) = (-4) \times 2$
2.	$6 \times (-5) =$	$(-5) \times 6 =$	
3.	$(-7) \times 6 =$	$6 \times (-7) =$	
4.	$(-9) \times (-8) =$	$(-8) \times (-9) =$	
5.	$(-6) \times (-9) =$	$(-9) \times (-6) =$	
6.	$11 \times 7 =$	$7 \times 11 =$	
7.	$(-13) \times 0 =$	$0 \times (-13) =$	$(-13) \times 0 = 0 \times (-13)$
8.	$(-8) \times 1 =$	$1 \times (-8) =$	

(i) What do you observe from the above table?

(ii) Is $a \times (-b) = (-b) \times a$?

Give examples to support your answer.

HOTS QUESTIONS



- Find an integer which when multiplied by -1 and divided by -9 results into reciprocal of that integer.
- A railway engine starts shunting wagons by first going 1000 m north of railway station. It reverses and travels 1250 m south and then it again moves 600 m north. Finally moving 800 m south it comes to rest. How far is the engine from the station and in which direction?

ANSWERS



Exercise 1.1

- (i) -56 (ii) -455 (iii) 173 (iv) 17 (v) -196
- (i) -88 (ii) -7 (iii) 135 (iv) 32 (v) 69 (vi) 85

- 3 , -246
- 4 , 368
- 5 , -122
- 6 , 321

Exercise 1.2

- (i) -2 (ii) -56
- (i) 50 (ii) -49

- (i) -72 (ii) 60 (iii) 85 (iv) 70 (v) -300 (vi) 0
- (i) 12300 (ii) 720 (iii) -120 (iv) 36

- (i) 3276 (ii) 896400 (iii) 678000 (iv) 23140

- (i) $\frac{1}{6}$ (ii) -3 (iii) $-\frac{4}{1}$ (iv) $-\frac{2}{5}$

- (i) 5700 (ii) -3100 (iii) 19400 (iv) 48960

- (i) 3 (ii) 34 (iii) 97

Exercise 1.3

- (i) -5 (ii) -27 (iii) 9 (iv) 25 (v) -144

- (i) 81 (ii) 7 (iii) 6 (iv) -5 (v) -30

- (i) -21 (ii) 15 (iii) -24 (iv) 14

- (i) 1 (ii) -8 (iii) 9 (iv) 6
- (i) -2 (ii) -8 (iii) 12 (iv) -3 (v) -12 (vi) 3

- $\frac{3}{1}$, $\frac{5}{3}$, $-\frac{4}{7}$

Exercise 1.4

- -15
- -63
- 72
- 150

Multiple Choice Questions

- (a)
- (b)
- (d)
- (b)
- (a)
- (b)
- (a)
- (c)
- (a)
- (b)
- (d)
- (b)
- (b)
- (c)

Mental Maths Corner

- Positive integer
- negative

- Positive

- -174
- -7
- $a - 1$
- $9 - 150$
- $0, 1$
- smaller

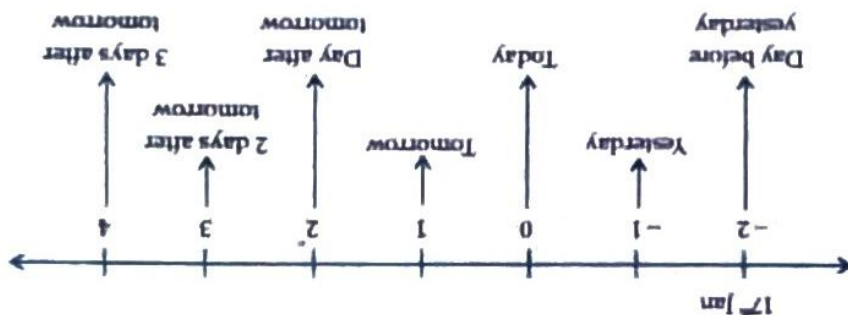
Review Exercise

- 2
- (i) $(6 - 8) \times 10$ (ii) $(12 + 6) \times 10$
- (i) $(-3) \times (-7) \times (-2)$
- (i) 2396700 (ii) 776000 (iii) 376400
- (i) $(-8, 3)$ (ii) $(2, 11)$ (iii) $(6, -6)$ (iv) $(3, -5)$
- (i) $(3, 8)$ (ii) $(12, 2)$ (iii) $(-21, 3)$
- 5600 m
- No
- (i) -31 (ii) 0
- (i) Negative (ii) Negative (iii) Negative (iv) Positive

HOTS Questions

- 3
- 450 m south

Hence, the date 3 days after tomorrow is 23rd January.



Solution :

Example 4 :

Taking today as zero on the number line, if the day before yesterday is 17 January, what is the date 3 days after tomorrow?

Net balance remaining after payment = ₹ 850 - ₹ 360 = ₹ 490

Payment done = ₹ 120 + ₹ 240 = ₹ 360

= ₹ 500 + ₹ 350 = ₹ 850

Hence, the total amount = previous balance + deposited amount

Cash deposited = ₹ 200 + ₹ 150 = ₹ 350

Solution :

Cheque no.	Date	Transaction	Payment	Deposit
384102	4/9	Jal Board Deposit	₹ 120	₹ 200
275146	12/9	Deposit		
384103	22/9	LIC India Deposit	₹ 240	
801351	29/9	Deposit		₹ 150

money is in your account after these transactions?

Example 3 :

You have ₹ 500 in your savings account at the beginning of the month. The record below shows all of your transactions during the month. How much

well = New water level + Height of wall + Height of pulley

Now, the minimum length of the rope that he can use to draw water from the

= 200 cm = 2 m

= 120 cm + 80 cm

Therefore, total height above ground level = Height of wall + Height of pulley

80 cm.

Wall of the well is 1 m 20 cm = 120 cm high and pulley is fixed at a height of

New water level = - 20 m + 5 m = - 15 m

Due to rain harvesting its level increases by 5 m.

Initial water level in the well = - 20 m

[∴ It is below ground level]

We have

Solution :

Example 1 : In an objective type test containing 25 questions, a student is to be awarded +5 marks for every correct answer, -5 for every incorrect answer and zero for not writing any answer. Mention the ways of scoring 110 marks by a student.

Solution : Marks that student has got = 110

$$\therefore \text{Minimum correct responses must be } \frac{110}{5} = 22$$

There are two cases in which student can score 110.

By giving 22 correct responses.

Each response gives him 5 marks

$$\text{Hence, total marks} = 22 \times 5 = 110$$

No incorrect answer will give $-5 \times 0 = 0$ marks

Unattempted three questions will give $3 \times 0 = 0$ marks

$$\therefore \text{Total marks} = 110 + 0 + 0 = 110$$

By giving 23 correct responses

Each response gives him 5 marks

$$\text{Hence, total marks} = 23 \times 5 = 115$$

Each incorrect response will give -5 marks

$$\text{Hence, the marks obtained by 1 wrong answer} = -5 \times 1 = -5$$

Unattempted question will give 0 marks

$$\text{Hence, total marks} = 115 + (-5) + 0$$

$$= 115 - 5 = 110$$

Example 2 :

Water level in a well was 20 m below ground level. During rainy season, rain water collected in different water tanks was drained into the well and the water level rises 5 m above the previous level. The wall of the well is 1 m 20 cm high and a pulley is fixed at a height of 80 cm. Raghu wants to draw water from the well. Find the minimum length of the rope that he can use.

